

Multimodal AI

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Multimodal AI

= AI working with more than one type of data (modality)



Text



Speech and other audio



Images



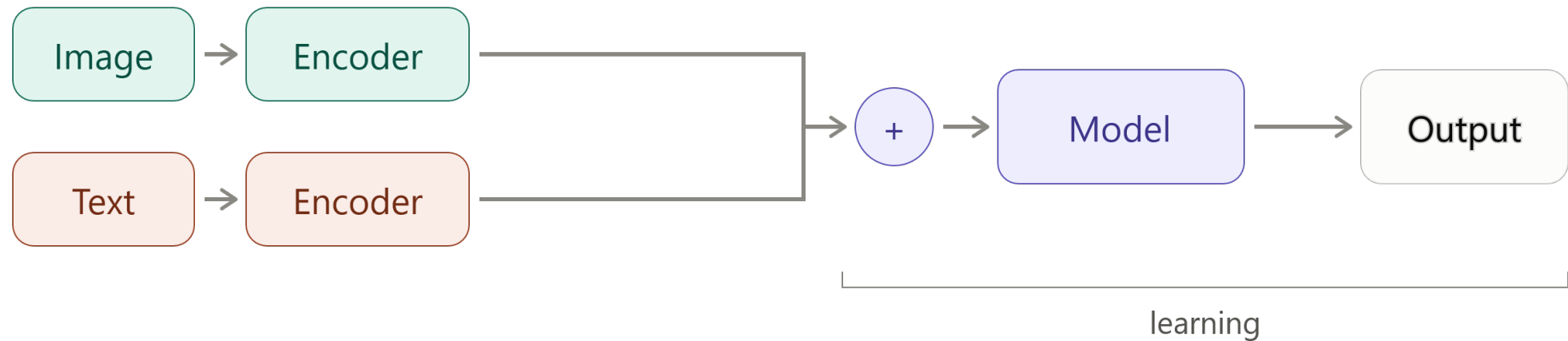
Video



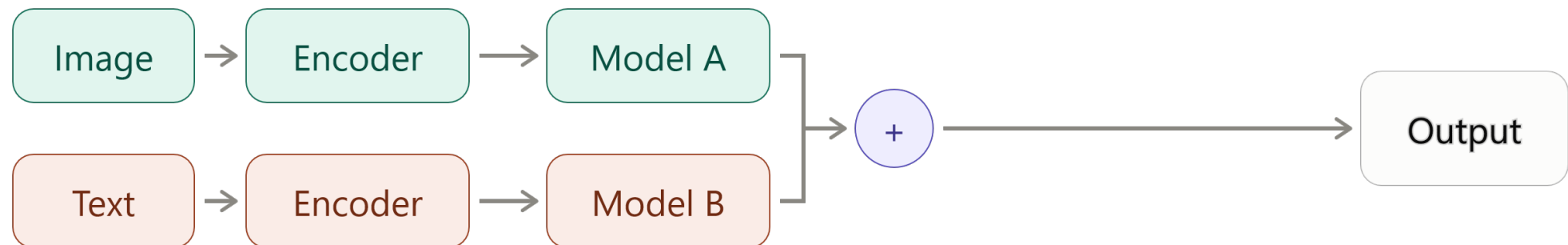
Structured data — e.g. sensor measurements, omics, lab results

Multimodal data integration

Early fusion



Late fusion



Produce a simple image that illustrates the concept of multimodal AI.

Image
generation

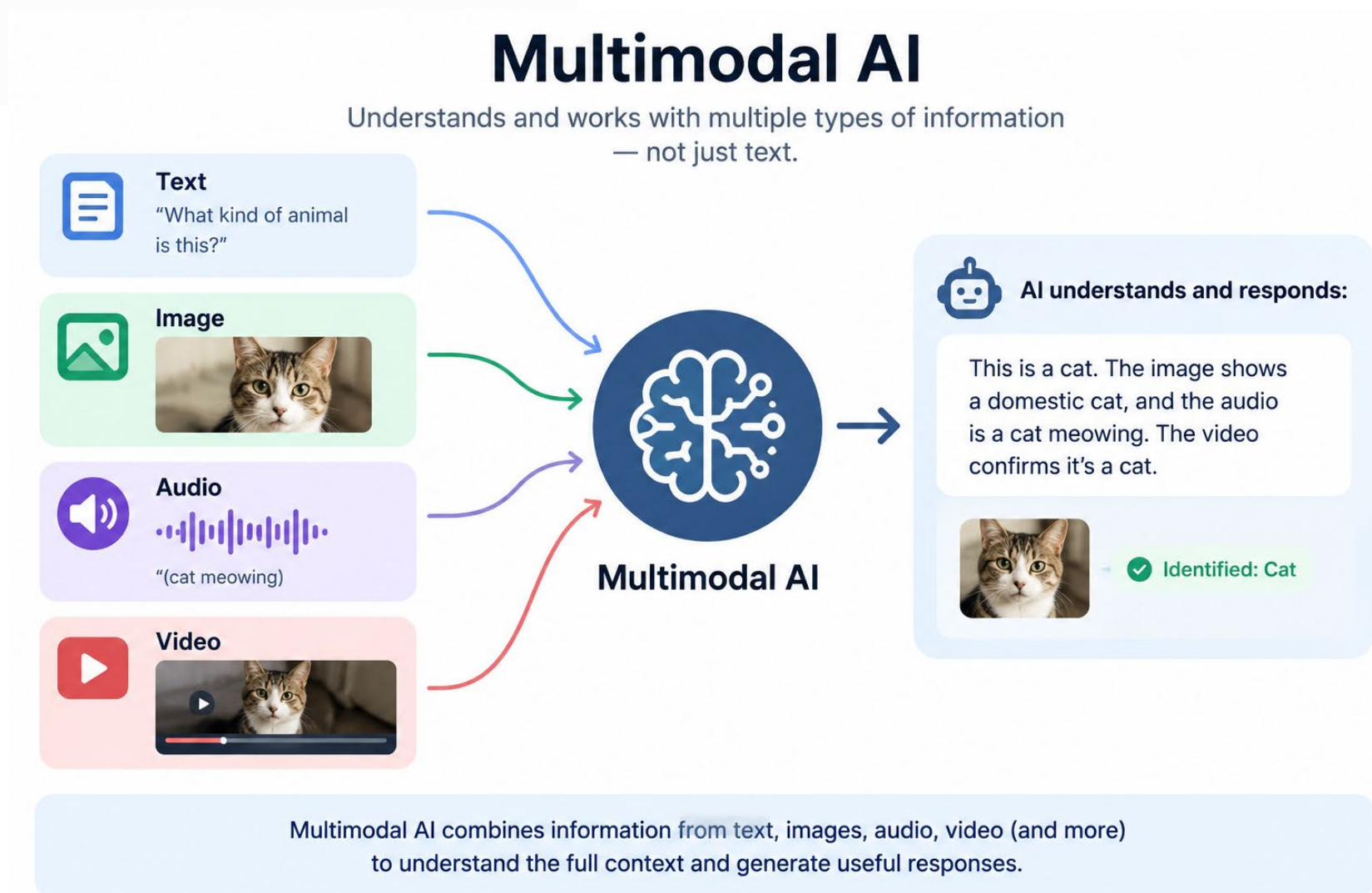
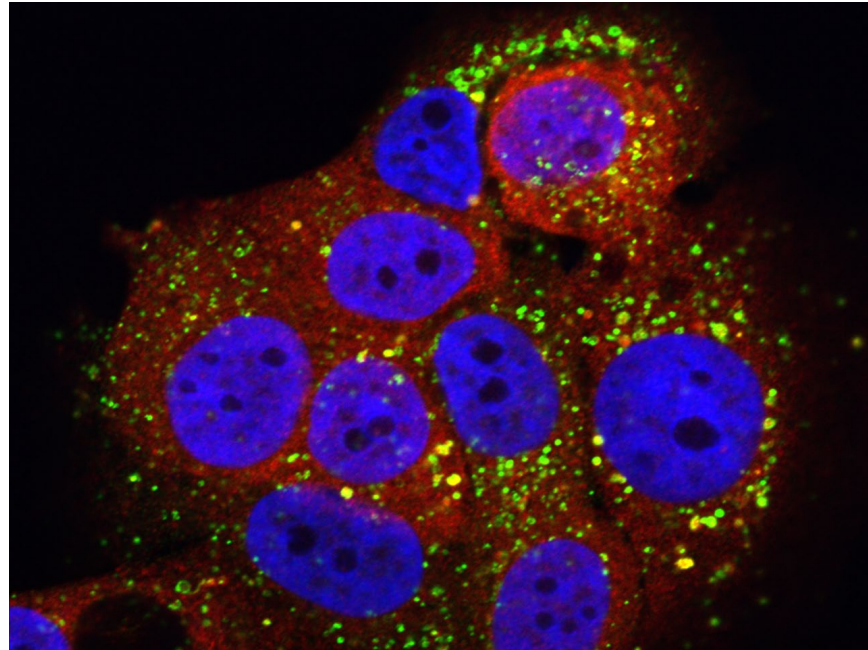
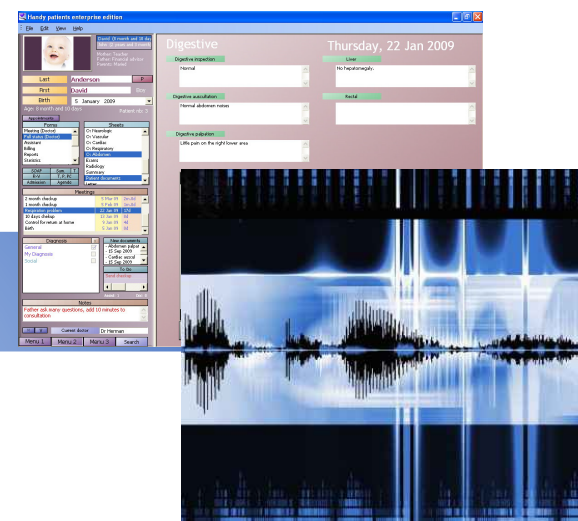
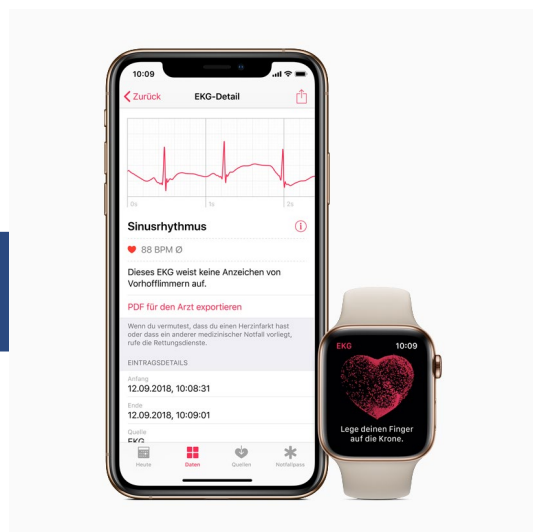


Image captioning



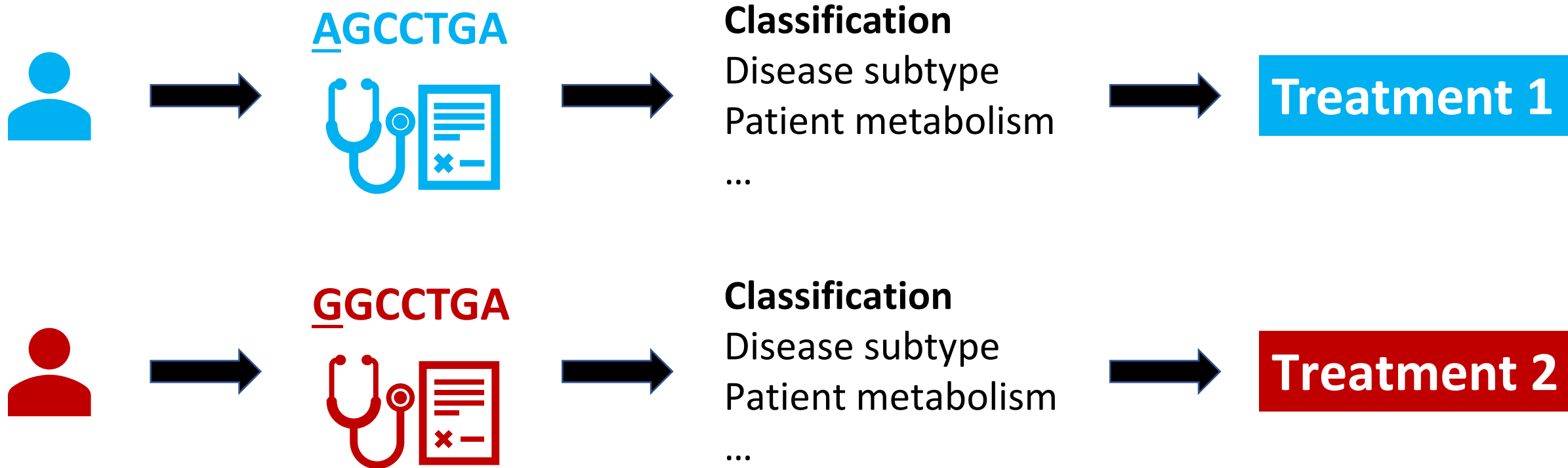
“a fluorescence microscopy image showing a group of cells with blue nuclei and bright green cytosolic spots”

Multimodal collection of patient history



**Comprehensive
patient history**

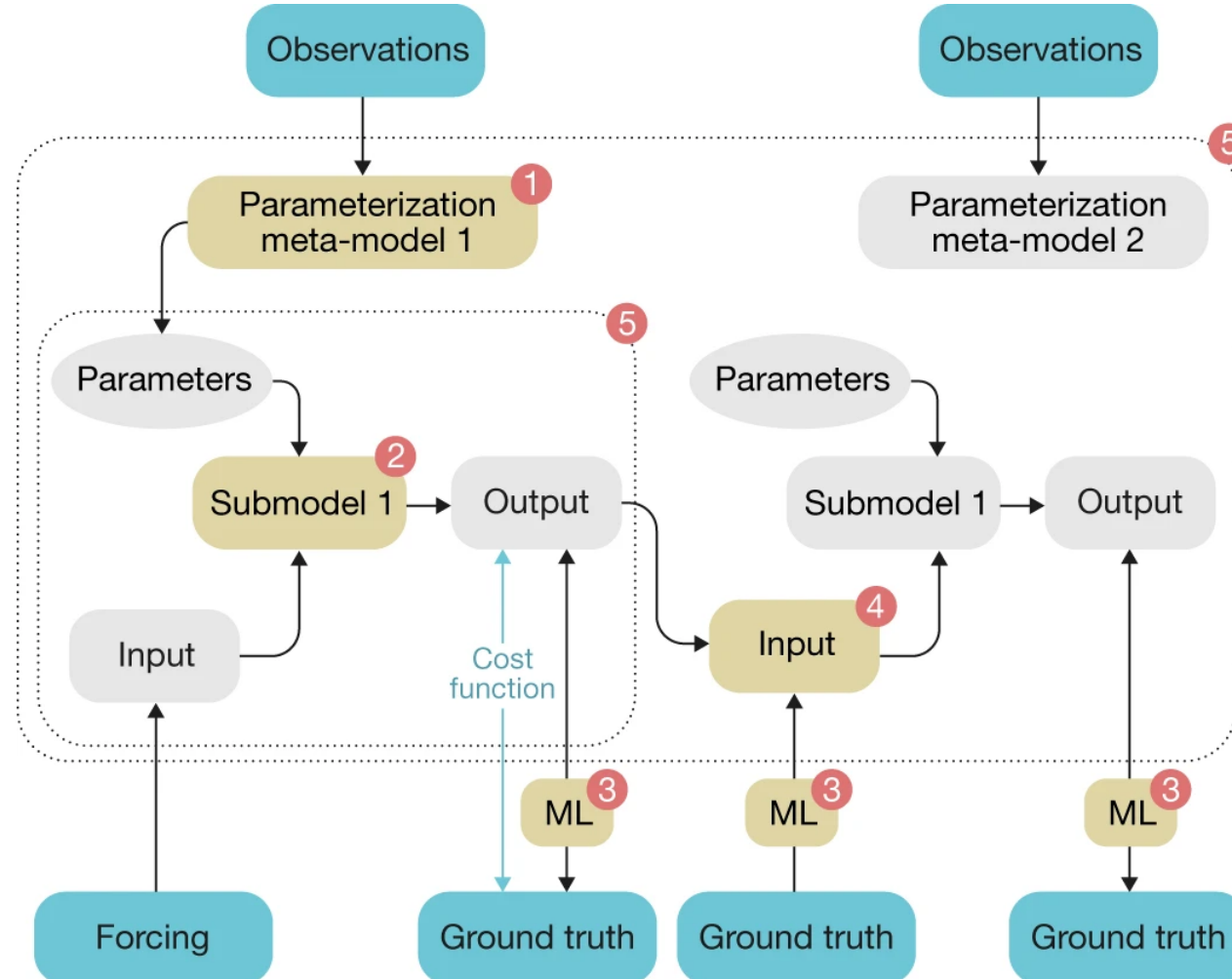
AI-based predictions for personalized medicine rely on multi-modal data



Computer vision-driven assistant devices



AI can enhance “traditional” modelling



Summary

- Complex tasks often require a combination of different data types → multimodal AI
- Complex AI systems can also combine multiple types of AI as well as non-AI aspects